

LESHENG LI

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CORE PROFICIENCIES

Computational Chemist and Materials Scientist with 10+ years leading atomic/multi-scale modeling and simulations for semiconductor and optoelectronic devices. Expert in gas phase and surface reactions, DFT/MD-to-KMC integration, UMLIP automation, and PINN-based PDE solvers. Drives semiconductor processes (CVD/ALD/etch) optimization and delivers simulation-powered breakthroughs for next generation device innovation.

TECHNICAL SKILLS

- **Simulation and Modeling:** First-principles simulations (e.g., DFT), Molecular Dynamics (MD), Kinetic Monte Carlo (KMC)
- **Machine Learning Frameworks:** Universal Machine-Learning Interatomic Potential (UMLIP) automation workflow and Physics-Informed Neural Networks (PINN) based PDE solver development
- **Process Technologies:** Epitaxy, ALD, chemical/thermal/sputter etch, particle diffusion, gas phase and surface reactions
- **Software and Programming:** Extensive experience with simulation software like LAMMPS, VASP, Quantum ESPRESSO, etc.; in-house code development and maintenance; proficiency in Python for multi-scale modeling and automation workflow
- **Management and Collaboration:** Team leadership in method development, protocol definition, cross-functional collaboration

PROFESSIONAL APPOINTMENTS

- **Technical Manager**
TSMC Technology Inc., R&D TCAD
June 2025 – Present
- **Principal Engineer**
TSMC Technology Inc., R&D TCAD
July 2021 – June 2025
- **Postdoctoral Researcher**
Princeton University, Supervisor: Prof. Emily A. Carter
May 2018 – June 2021

EDUCATIONAL BACKGROUND

Ph.D.	University of North Carolina at Chapel Hill <i>Major: Chemistry (Theoretical and Computational Chemistry)</i>	Advisor: Prof. Yosuke Kanai	<i>April 2018</i>
M.S.	Xi'an Jiaotong University <i>Major: Materials Science and Engineering</i>	Advisor: Prof. Xiang Zhao	<i>June 2021</i>
B.S.	Xi'an Jiaotong University <i>Major: Applied Chemistry</i>		<i>June 2008</i>

RESEARCH EXPERIENCE

Technical Manager, TSMC Technology Inc., R&D TCAD

June 2025 – Present

- Lead a specialized team in developing state-of-the-art atomic- and continuum-level methods/tools, including gas phase reaction simulator, surface reaction simulator, Kinetic Monte Carlo simulator, Universal Machine-Learning Interatomic Potential (UMLIP) automation workflow, and PDE solvers using Physics-Informed Neural Networks (PINN).
- Direct delivery of reliable simulations to support experimental teams, enabling device optimization and providing actionable insights for refining key processes such as epitaxy, ALD, chemical/thermal/sputter etch, and particle diffusion.
- Foster team collaboration to achieve breakthroughs in semiconductor process control, resulting in enhanced efficiency and innovation in chip design.

Principal Engineer, TSMC Technology Inc., R&D TCAD

July 2021 – June 2025

- Development and maintenance of multi-scale modeling methods and tools
 - Developed methods and tools which bridge atomic-level simulations (such as first-principles and molecular dynamics) with higher-level simulators (like Kinetic Monte Carlo) to drive device optimization and chip design advancements.
 - Provided comprehensive training, assistance, and support to users of in-house codes, ensuring seamless integration into daily workflows.
- Multi-scale simulations of surface reactions for CVD/ALD/etch/diffusion
 - Investigated reaction mechanisms of precursors on various substrates using first-principles simulations and deep neural network (DNN)-assisted molecular dynamics.
 - Streamlined atomic simulation parameters into device-level models via Kinetic Monte Carlo, offering insights to fine-tune recipes and improve process outcomes.
- Detailed process control in cutting-edge process technology
 - Applied DNN-assisted molecular dynamics with enhanced sampling techniques to analyze competing chemical reactions in cutting-edge process technologies, deriving novel mechanisms to inform experimental strategies.
 - Delivered recommendations to experimental teams, facilitating breakthroughs and optimizing device performance.
- In-house Machine-Learning Interatomic Potential (MLIP) database construction
 - Led the construction of an in-house MLIP database, establishing protocols for training and validation of MLIPs across bulk materials, surfaces, and interfaces.
 - Supervised internal engineers in generating accurate MLIPs, validated results, created benchmarking procedures, and maintained the in-house MLIP database for ongoing R&D support.

Postdoctoral Researcher, Princeton University

May 2018 – June 2021

- Water purification through electrochemical desalination
 - Combined deep neural networks and first-principles theory to accelerate materials discovery for identifying novel materials as potential capacitive deionization (CDI) electrodes.
 - Investigated ion and water intercalation mechanisms into energy-storage materials as CDI electrodes for water desalination.
- Heterogeneous catalysis for valuable fuel generation
 - Artificial ammonia synthesis via electrocatalytic nitrogen reduction.
 - Photoelectrochemical reduction of CO₂ for solar fuel generation.
- Quantum dynamics of excited carriers in low-dimensional photoactive materials
 - Investigated electron-phonon interactions and nonadiabatic effects in transition metal dichalcogenide (WSe₂) monolayers for optoelectronic device applications.

SELECTED PUBLICATIONS

1. N. Xu, L. Li, K.C. Ong, H.-L. Lin, Z.-R. Xiao, Q. Shi, Y.-H. Chang, J. Wu, "GPU Accelerated Semiconductor Simulations: from Atom to Transistor", **NVIDIA GTC**, 2025.
2. L. Li, M. F. Calegari, Andrade, A. Selloni, R. Car, and E. A. Carter, "Characterizing Structure-Dependent TiS₂/H₂O Interfaces using Deep-Neural-Network-Assisted Molecular Dynamics", **J. Phys. Chem. C** 2023, 127, 9750.
3. L. Li, M. Agrawal, S. Y. Yeh, K. T. Lam, J. Wu, and B. Magyari-Köpe, "Towards Accurate and Efficient Process Simulations based on Atomistic and Neural Network Approaches", **2022 International Electron Devices Meeting (IEDM)**, San Francisco, CA, USA, 2022, pp. 15.6.1-15.6.4, doi: 10.1109/IEDM45625.2022.10019371.
4. L. Li, J. M. P. Martinez, and E. A. Carter, "Identifying an Alternative Hydride Transfer Pathway for CO₂ Reduction on CdTe(111) and CuInS₂(112) Surfaces". **Adv. Theory Simul.** 2022, 5, 2100413.
5. L. Li, S. Xu, and E. A. Carter, "First-Principles Modeling of Sodium Ion and Water Intercalation into Titanium Disulfide Interlayers for Water Desalination". **Chem. Mater.** 2020, 32, 10678.
6. L. Li, J. M. P. Martinez, and E. A. Carter, "Prediction of Highly Selective Electrocatalytic Nitrogen Reduction at Low Overpotential on a Mo-doped g-GaN Monolayer". **ACS Catal.** 2020, 10, 12841.
7. A. Taggart, J. Evans, L. Li, J. Lee, L. Dempsey, Y. Kanai, and J. Cahoon, "Enabling Aqueous NiO Photocathodes by Passivating Surface Sites that Facilitate Proton-Coupled Charge Transfer". **ACS Appl. Energy Mater.** 2020, 3, 10702.
8. L. Li and E. A. Carter, "Defect-Mediated Charge-Carrier Trapping and Nonradiative Recombination in WSe₂ Monolayer". **J. Am. Chem. Soc.** 2019, 141, 10451.
9. L. Li and Y. Kanai, "Modeling Electron Injection at Semiconductor-Molecule Interfaces using First-Principles Dynamics Simulation: Effects of Nonadiabatic Coupling, Self-Energy, and Surface Models". **J. Phys. Chem. C** 2019, 123, 13295.
10. S. Xu, L. Li, and E. A. Carter, "Why and How Carbon Dioxide Conversion to Methanol Happens on Functionalized Semiconductor Photoelectrodes". **J. Am. Chem. Soc.** 2018, 140, 16749.
11. L. Li and Y. Kanai, "Modeling Plasmon-Induced Hot-Carrier Transfer". **Chem** 2018, 4, 931.
12. J. C. Wong, L. Li, and Y. Kanai, "Size Dependence and Role of Decoherence in Hot Electron Relaxation within Fluorinated Silicon Quantum Dots: A First-Principles Study". **J. Phys. Chem. C** 2018, 122, 29526.
13. L. Li and Y. Kanai, "Dependence of Hot Electron Transfer on Surface Coverage and Adsorbate Species at Semiconductor-Molecule Interfaces". **Phys. Chem. Chem. Phys.** 2018, 20, 12986.
14. L. Li, J. C. Wong, and Y. Kanai, "Examining the Effect of Exchange-Correlation Approximations in First-Principles Dynamics Simulation of Interfacial Charge Transfer". **J. Chem. Theory Comput.** 2017, 13, 2634.
15. C. J. Flynn, S. M. McCullough, L. Li, C. L. Donley, Y. Kanai, and J. F. Cahoon, "Passivation of Nickel Vacancy Defects in Nickel Oxide Solar Cells by Targeted Atomic Deposition of Boron". **J. Phys. Chem. C** 2016, 120, 16568.
16. L. Li and Y. Kanai, "Excited Electron Dynamics at Semiconductor-Molecule Type-II Heterojunction Interface: First-Principles Dynamics Simulations". **J. Phys. Chem. Lett.** 2016, 7, 1495.
17. C. J. Flynn, S. M. McCullough, E. B. Oh, L. Li, C. C. Mercado, B. H. Farnum, W. Li, C. L. Donley, W. You, A. J. Nozik, J. R. McBride, T. J. Meyer, Y. Kanai, and J. F. Cahoon, "Site-Selective Passivation of Defects in NiO Solar Photocathodes by Targeted Atomic Deposition". **ACS Appl. Mater. Interfaces** 2016, 8, 4754.
18. L. Li and Y. Kanai, "Antiferromagnetic Structures and Electronic Energy Levels at Reconstructed NiO(111) Surfaces: A DFT+U Study". **Phys. Rev. B** 2015, 91, 235304.
19. L. Li, P. G. Giokas, Y. Kanai, and A. M. Moran, "Modeling Time-coincident Ultrafast Electron Transfer and Solvation Processes at Molecule-Semiconductor Interfaces". **J. Chem. Phys.** 2014, 140, 234109.
20. T. Yang, X. Zhao, L. Li, J. Zheng, and W. Gao, "Large Gadolinium Nitride Cluster Encapsulated Inside a Non-IPR Carbon Cage: A Theoretical Characterization on Gd₃N@C₇₈". **ChemPhysChem** 2012, 13, 449.
21. X. Zhao, W. Gao, T. Yang, J. Zheng, L. Li, L. He, R. Cao, and S. Nagase, "Violating the Isolated Pentagon Rule (IPR): Endohedral Non-IPR C₉₈ Cages of Gd₂@C₉₈". **Inorg. Chem.** 2012, 51, 2039.
22. L. Li and X. Zhao, "Transformation Between Different Hybridized Bonding Structures in Two-dimensional Diamond-based Materials". **J. Phys. Chem. C** 2011, 115, 22168.
23. L. Li and X. Zhao, "Dangling Bond-induced Graphitization Process on the (111) Surface of Diamond Nanoparticles". **J. Chem. Phys.** 2011, 134, 044711.

PROFESSIONAL ACTIVITIES

- **Referee Service for**
Angewandte Chemie, The Journal of Physical Chemistry Letters, The Journal of Physical Chemistry A/C, ACS Physical Chemistry Au, ChemSusChem, Chemical Physics Letters
- **Membership of**
American Chemical Society, American Physical Society, American Institute of Chemical Engineers